Experiment 3

Energy and Power Analysis of Signals

OBJECTIVE:

To calculate the energy and average power of given continuous-time and discrete time signals and classify them as energy signals, power signals, or neither based on their energy and power characteristics.

Procedure :

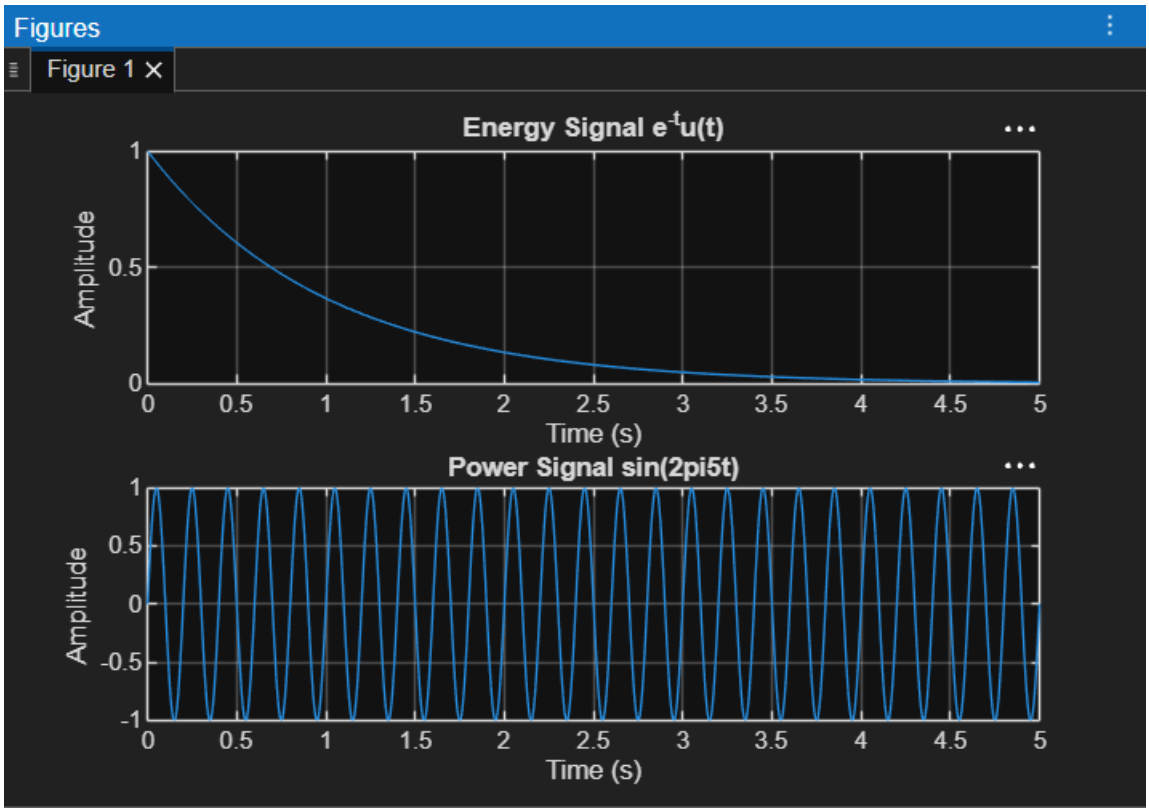
1. Define the time vector and generate the selected signals.
2. Compute the squared magnitude of each signal.
3. Numerically integrate over a sufficiently large interval for energy.
4. Compute average power over an interval containing several periods for the periodic signal.
5. Compare numerical results with analytical expectations.
6. Classify the signals and record observations.

MATLAB Coding

```
% Experiment 3: Energy and Power Analysis
clc; clear; close all;
dt = 0.001;
t = 0:dt:5;
xE = exp(-t);
xP = sin(2*pi*5*t);
E_num = trapz(t,abs(xE).^2);
T0 = 1/5;
idx = t <= 2;
P_num = (1/(t(find(idx,1,'last'))-t(1))) * ...
trapz(t(idx),abs(xP(idx)).^2);
figure;
subplot(2,1,1);
```

```
plot(t,xE); grid on;
title('Energy Signal e^{-t}u(t)');
xlabel('Time (s)'); ylabel('Amplitude');
subplot(2,1,2);
plot(t,xP); grid on;
title('Power Signal sin(2pi5t)');
xlabel('Time (s)'); ylabel('Amplitude');
fprintf('Numerical energy = %g\n',E_num);
fprintf('Numerical average power = %g\n',P_num);
```

OUTPUT:



RESULT:

Two discrete-time sequences were generated and their addition and point-wise multiplication

were performed successfully using MATLAB. The resulting sequences were displayed using

stem plots, illustrating basic operations on discrete-time signals.

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
1. Generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal and exponential.	**CT Signal Generation & MATLAB Implementation**	Generates all five CT signals accurately with a suitable time vector and step size. Uses correct MATLAB commands . Plots are correctly scaled, with titles, axis labels and grid.	Generates the required CT signals and plots correctly, with minor errors in code, time range, parameters or labelling.	Generates basic CT signals but has noticeable errors in code, step or impulse representation, or plots, and needs assistance.	Unable to generate the required CT signals or produce meaningful plots.	**25**	**2.5**	
2. Generate and visualize basic discrete-time signals such as unit sample, unit step, ramp, sinusoidal and exponential sequences.	**DT Signal Generation & MATLAB Implementation**	Generates all five DT sequences accurately with a proper index vector n. Uses stem correctly. Plots are correctly scaled, with titles, axis labels and grid.	Generates the required DT sequences and plots correctly, with minor errors in index range, stem usage, parameters or labelling.	Generates basic DT sequences but has noticeable errors in code, indexing or stem plots, and needs assistance.	Unable to generate the required DT sequences or produce meaningful stem plots.	**25**	**2.5**	
3. Classify CT and DT signals as periodic or aperiodic and identify their waveform characteristics.	**Signal Classification & Periodicity Analysis**	Correctly classifies all signals using DT. Determines the period where applicable. Identifies frequency, phase, and growth or decay. Explains the	Classifies most signals correctly and finds most periods, with minor errors or limited justification.	Gives basic classifications . Period calculation or justification is incomplete and needs guidance.	Unable to classify the signals correctly or justify the classification .	**25**	**2.5**	

		reasoning clearly with mathematical conditions and waveform evidence.						
4. Analyze the amplitude and time-index characteristics of discrete-time signals.	**Amplitude & Time-Index Analysis**	Correctly determines maximum and minimum amplitude, the index n at which each occurs, and signal energy. Correctly shows the effect of time shifting and folding on the index. Interprets the results clearly.	Analyzes most amplitude and index characteristics correctly, with minor errors in values or interpretation.	Gives basic amplitude and index analysis with noticeable errors, and needs guidance.	Unable to analyze the amplitude and time-index characteristics or interpret the results.	**25**	**2.5**	
					Total	**100**	**10**	